

离散数学 习题课

1. 考虑如图所示的十六进制7段数码管.

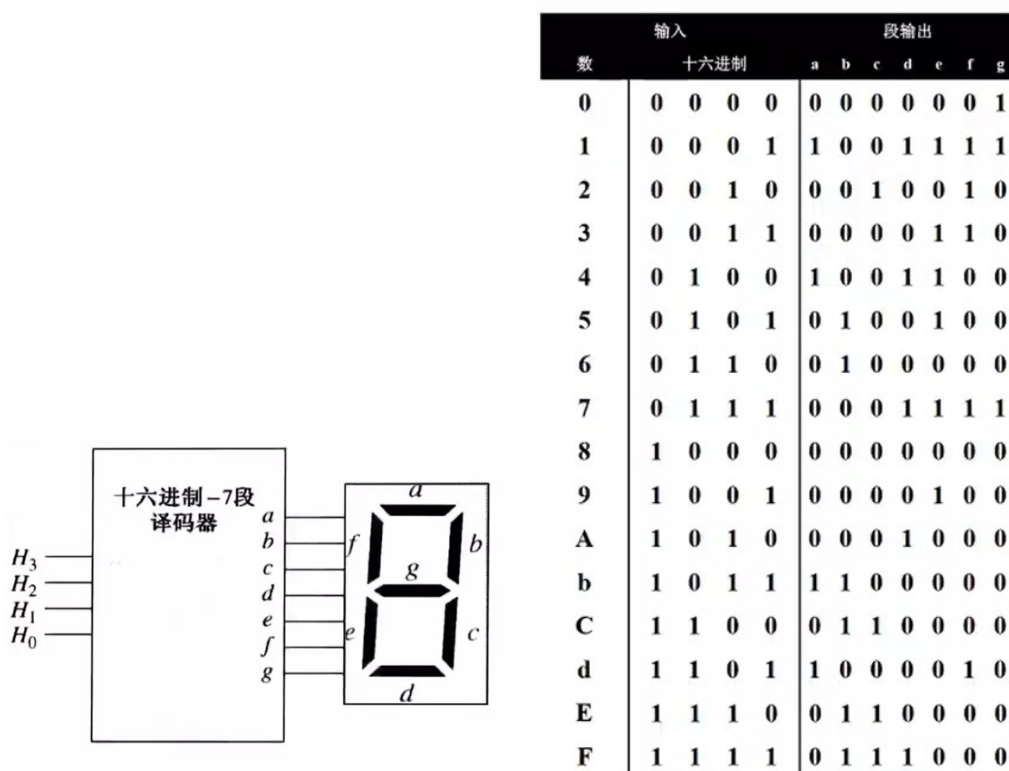


图 1: 第1题图

- 请利用卡诺图得到管b和管e的最简表示 (即表示为乘积项之和, 使总项数最少, 且每项中的因子也最少);
 - 如果只考虑用此7段数码管表示数字0 ~ 9, 而不表示字母a ~ f, 管b和管e的最简表示会发生变化吗?
- 一个国际社团的成员来自6个国家, 共有成员1978名, 用1, 2, ..., 1978给每个人编号. 试证明: 该社团中至少有一名成员, 他的编号等于他的两个同胞的编号之和, 或者是他的一个同胞的编号的两倍.
 - 设 $F(1) = 1, F(2) = 1, F(n) = F(n-1) + F(n-2) (n \geq 3)$. 证明: $\gcd(F(m), F(n)) = F(\gcd(m, n))$. 其中, $\gcd(m, n)$ 为 m, n 的最大公约数.

4. Find a generating function $S(x)$ such that

$$[x^n]S(x) = \sum_{k=0}^n \binom{r}{k} \binom{r}{n-2k}$$

5. Solve the recurrence

$$g_0 = 1$$

$$g_n = g_{n-1} + 2g_{n-2} + \cdots + ng_0, \quad \text{for } n > 0.$$

6. (a) What is $[x^n](\ln(1-x))^2/(1-x)^{m+1}$?

(b) Use the previous result to evaluate $\sum_{k=0}^n H_k H_{n-k}$. Here H_k stands for the Harmonic series.

7. We assume that $A(x) = \sum_n a_n x^n$, $B(x) = \sum_n b_n x^n$, $C(x) = \sum_n c_n x^n$, and the coefficients are zero for negative n .

(a) If $c_n = \sum_{j+2k \leq n} a_j b_k$, express C in terms of A and B .

(b) If $nb_n = \sum_{k=0}^n 2^k a_k / (n-k)!$, express A in terms of B .

(c) If r is a real number and if $a_n = \sum_{k=0}^n \binom{r+k}{k} b_{n-k}$, express A in terms of B ; then use your formula to find coefficients $f_k(r)$ such that $b_n = \sum_{k=0}^n f_k(r) a_{n-k}$.

8. In how many ways can a $2 \times 2 \times n$ pillar be built out of $2 \times 1 \times 1$ bricks?

9. What is the sum of Fibonacci products

$$\sum_{m>0} \sum_{\substack{k_1+k_2+\cdots+k_m=n \\ k_1, k_2, \dots, k_m > 0}} F_{k_1} F_{k_2} \cdots F_{k_m}?$$

10. Given positive integers m and n , find closed forms for

$$\sum_{1 \leq k_1 < k_2 < \cdots < k_m \leq n} k_1 k_2 \cdots k_m \quad \text{and} \quad \sum_{1 \leq k_1 \leq k_2 \leq \cdots \leq k_m \leq n} k_1 k_2 \cdots k_m.$$

(Hint: What are the coefficients of x^m in the generating function $(1+a_1x) \cdots (1+a_nx)$ and $1/(1-a_1x) \cdots (1-a_nx)$?)

11. 在1和100000之间有多少个整数包含了数字1, 2, 3, 4?

12. We obtain the formulas

$$\sum_{k \leq m} \binom{n+k}{k} = \binom{n+m+1}{m}, \quad \sum_{0 \leq k \leq m} \binom{k}{n} = \binom{m+1}{n+1}$$

from $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$. Then what can we get from the recurrence $\{n\}_k = k \{n-1\}_k + \{n-1\}_{k-1}$?

13. What are the closed forms of $\left[\begin{smallmatrix} n \\ 2 \end{smallmatrix} \right]$, $\left[\begin{smallmatrix} n \\ 3 \end{smallmatrix} \right]$, $\{n\}_2$, $\{n\}_3$?

14. Let P be the sum of all ways to “triangulate” polygons: Define a “multiplication” op-

$$P = \text{---} + \triangle + \square + \begin{array}{|c|} \hline \diagup \\ \hline \end{array} + \begin{array}{|c|} \hline \diagdown \\ \hline \end{array} \\ + \begin{array}{|c|} \hline \diagup \diagdown \\ \hline \end{array} + \begin{array}{|c|} \hline \diagdown \diagup \\ \hline \end{array} + \begin{array}{|c|} \hline \diagup \diagdown \diagup \\ \hline \end{array} + \begin{array}{|c|} \hline \diagdown \diagup \diagdown \\ \hline \end{array} + \begin{array}{|c|} \hline \diagup \diagdown \diagup \diagdown \\ \hline \end{array} + \cdots$$

图 2: 第14题图

eration $A \triangle B$ on triangulated polygons A and B so that the following equation is valid.

$$P = \text{---} + P \triangle P$$

Then replace each triangle by z . What does this tell you about the number of ways to decompose an n -gon into triangles?

15. Consider the sum of all ways to decompose polygons into polygons: Find a symbolic

$$Q = \text{---} + \triangle + \square + \begin{array}{|c|} \hline \diagup \\ \hline \end{array} + \begin{array}{|c|} \hline \diagdown \\ \hline \end{array} \\ + \begin{array}{|c|} \hline \diagup \diagdown \\ \hline \end{array} + \begin{array}{|c|} \hline \diagdown \diagup \\ \hline \end{array} + \begin{array}{|c|} \hline \diagup \diagdown \diagup \\ \hline \end{array} + \begin{array}{|c|} \hline \diagdown \diagup \diagdown \\ \hline \end{array} + \begin{array}{|c|} \hline \diagup \diagdown \diagup \diagdown \\ \hline \end{array} + \cdots$$

图 3: 第15题图

equation for Q and use it to find a generating function for the number of ways to draw nonintersecting diagonals inside a convex n -gon (Give a closed form for the generating function as a function of z ; you need not find a closed form for the coefficients.)

16. (中国余数定理) 设 m, n 为两个互质的正整数, a, b 是满足 $0 \leq a \leq m-1, 0 \leq b \leq n-1$ 的整数. 证明: 存在正整数 x , 使得 x 除以 m 的余数为 a , 除以 n 的余数为 b , 即存在 p, q , 使得

$$x = pm + a, \quad x = qn + b.$$

17. 欧拉函数 $\phi(n)$ 表示小于 n 且与 n 互质的整数的个数. 求 $\phi(n)$.
18. 将两个相同的白球和两个相同的黑球放入两个不同的盒子里, 问有多少种不同的放法? 列出全部方案. 又问每个盒中有两个球的放法有多少种?

$$\begin{aligned}\alpha F(z) + \beta G(z) &= \sum_n (\alpha f_n + \beta g_n) z^n \\ z^m G(z) &= \sum_n g_{n-m} z^n, \quad \text{integer } m \geq 0 \\ \frac{G(z) - g_0 - g_1 z - \cdots - g_{m-1} z^{m-1}}{z^m} &= \sum_{n \geq 0} g_{n+m} z^n, \quad \text{integer } m \geq 0 \\ G(cz) &= \sum_n c^n g_n z^n \\ G'(z) &= \sum_n (n+1) g_{n+1} z^n \\ zG'(z) &= \sum_n n g_n z^n \\ \int_0^z G(t) dt &= \sum_{n \geq 1} \frac{1}{n} g_{n-1} z^n \\ F(z)G(z) &= \sum_n \left(\sum_k f_k g_{n-k} \right) z^n \\ \frac{1}{1-z} G(z) &= \sum_n \left(\sum_{k \leq n} g_k \right) z^n\end{aligned}$$

图 4: 生成函数的操作

sequence	generating function	closed form
$\langle 1, 0, 0, 0, 0, 0, \dots \rangle$	$\sum_{n \geq 0} [n=0] z^n$	1
$\langle 0, \dots, 0, 1, 0, 0, \dots \rangle$	$\sum_{n \geq 0} [n=m] z^n$	z^m
$\langle 1, 1, 1, 1, 1, 1, \dots \rangle$	$\sum_{n \geq 0} z^n$	$\frac{1}{1-z}$
$\langle 1, -1, 1, -1, 1, -1, \dots \rangle$	$\sum_{n \geq 0} (-1)^n z^n$	$\frac{1}{1+z}$
$\langle 1, 0, 1, 0, 1, 0, \dots \rangle$	$\sum_{n \geq 0} [2 \setminus n] z^n$	$\frac{1}{1-z^2}$
$\langle 1, 0, \dots, 0, 1, 0, \dots, 0, 1, 0, \dots \rangle$	$\sum_{n \geq 0} [m \setminus n] z^n$	$\frac{1}{1-z^m}$
$\langle 1, 2, 3, 4, 5, 6, \dots \rangle$	$\sum_{n \geq 0} (n+1) z^n$	$\frac{1}{(1-z)^2}$
$\langle 1, 2, 4, 8, 16, 32, \dots \rangle$	$\sum_{n \geq 0} 2^n z^n$	$\frac{1}{1-2z}$
$\langle 1, 4, 6, 4, 1, 0, 0, \dots \rangle$	$\sum_{n \geq 0} \binom{4}{n} z^n$	$(1+z)^4$
$\langle 1, c, \binom{c}{2}, \binom{c}{3}, \dots \rangle$	$\sum_{n \geq 0} \binom{c}{n} z^n$	$(1+z)^c$
$\langle 1, c, \binom{c+1}{2}, \binom{c+2}{3}, \dots \rangle$	$\sum_{n \geq 0} \binom{c+n-1}{n} z^n$	$\frac{1}{(1-z)^c}$
$\langle 1, c, c^2, c^3, \dots \rangle$	$\sum_{n \geq 0} c^n z^n$	$\frac{1}{1-cz}$
$\langle 1, \binom{m+1}{m}, \binom{m+2}{m}, \binom{m+3}{m}, \dots \rangle$	$\sum_{n \geq 0} \binom{m+n}{m} z^n$	$\frac{1}{(1-z)^{m+1}}$
$\langle 0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \rangle$	$\sum_{n \geq 1} \frac{1}{n} z^n$	$\ln \frac{1}{1-z}$
$\langle 0, 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots \rangle$	$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} z^n$	$\ln(1+z)$
$\langle 1, 1, \frac{1}{2}, \frac{1}{6}, \frac{1}{24}, \frac{1}{120}, \dots \rangle$	$\sum_{n \geq 0} \frac{1}{n!} z^n$	e^z

图 5: 一些数列与对应的生成函数

$$\begin{aligned}
\frac{1}{(1-z)^{m+1}} \ln \frac{1}{1-z} &= \sum_{n \geq 0} (H_{m+n} - H_m) \binom{m+n}{n} z^n \\
\frac{z}{e^z - 1} &= \sum_{n \geq 0} B_n \frac{z^n}{n!} \\
\frac{F_m z}{1 - (F_{m-1} + F_{m+1})z + (-1)^m z^2} &= \sum_{n \geq 0} F_{m,n} z^n \\
\sum_k \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{k! z^k}{(1-z)^{k+1}} &= \sum_{n \geq 0} n^m z^n \\
(z^{-1})^{-\overline{m}} &= \frac{z^m}{(1-z)(1-2z)\dots(1-mz)} = \sum_{n \geq 0} \left\{ \begin{matrix} n \\ m \end{matrix} \right\} z^n \\
z^{\overline{m}} &= z(z+1)\dots(z+m-1) = \sum_{n \geq 0} \left[\begin{matrix} m \\ n \end{matrix} \right] z^n \\
(e^z - 1)^m &= m! \sum_{n \geq 0} \left\{ \begin{matrix} n \\ m \end{matrix} \right\} \frac{z^n}{n!} \\
\left(\ln \frac{1}{1-z} \right)^m &= m! \sum_{n \geq 0} \left[\begin{matrix} n \\ m \end{matrix} \right] \frac{z^n}{n!} \\
\left(\frac{z}{\ln(1+z)} \right)^m &= \sum_{n \geq 0} \frac{z^n}{n!} \left\{ \begin{matrix} m \\ m-n \end{matrix} \right\} / \binom{m-1}{n} \\
\left(\frac{z}{1-e^{-z}} \right)^m &= \sum_{n \geq 0} \frac{z^n}{n!} \left[\begin{matrix} m \\ m-n \end{matrix} \right] / \binom{m-1}{n} \\
e^{z+wz} &= \sum_{m,n \geq 0} \binom{n}{m} w^m \frac{z^n}{n!} \\
e^{w(e^z-1)} &= \sum_{m,n \geq 0} \left\{ \begin{matrix} n \\ m \end{matrix} \right\} w^m \frac{z^n}{n!} \\
\frac{1}{(1-z)^w} &= \sum_{m,n \geq 0} \left[\begin{matrix} n \\ m \end{matrix} \right] w^m \frac{z^n}{n!}
\end{aligned}$$

图 6: 含特殊计数的生成函数